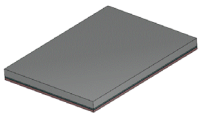
1/2.8-inch 2 MP CMOS Digital Image Sensor with Global Shutter AR0235CS



ODCSP62
CASE 570CY

ORDERING INFORMATION

See detailed ordering and shipping information on page 2 of this data sheet.

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Description

The AR0235CS is a 1/2.8-inch CMOS digital image sensor with an active-pixel array of 1920 (H) x 1200 (V). It incorporates a new innovative global shutter pixel design optimized for accurate and fast capture of moving scenes. The sensor produces clear, low noise images in both low-light and bright scenes. It includes sophisticated camera functions such as auto exposure control, windowing, row skip mode, column-skip mode, pixel-binning and both video and single frame modes. It is programmable through a simple two-wire serial interface. The AR0235CS produces extraordinarily clear, sharp digital pictures, and its ability to capture both continuous video and single frames makes it the perfect choice for a wide range of applications, including scanning and industrial inspection.

Table 1. KEY PERFORMANCE PARAMETERS

Parameter	Typical Value
Optical Format	1/2.8-inch (6.34 mm)
Active Pixels	1920 (H) x 1200 (V) not including 8 border pixels on each side
Pixel Size	2.8 μm
Color Filter Array	RGB Bayer, Monochrome
Chief Ray Angle	0 or 28°
Shutter Type	Global Shutter
Input Clock Range	10−48 MHz
Output Interface	8-bit/10-bit MIPI 1, 2, or 4-lane
Output Data Rate	Maximum Serial Output Data Rate 850 Mbps/lane
Frame Rate Full Resolution	120 fps (10-bit)
Responsivity	41.9 ke-/lux*s (Monochrome) TBD (RGB)
SNR_{MAX}	37 dB
Dynamic Range	65.3 dB
Supply Voltage I/O Digital Analog	 1.8 V 1.25 V 2.8 V
Power Consumption	252 mW, at 120 fps, Full Resolution
Operating Temperature	(−30 °C < T_J < +85 °C)
Optimal Performance Temperature Range	(0 °C < T_J < +60 °C)
Package Options	7.66 x 5.64 mm 62-ball CSP θ_{JA}: 32 °C/W (Note 1) θ_{JB}: 11 °C/W
	Bare Die

1. θ_{JA}" is dependent on the customer module design and should not be used for calculating junction temperature.

Features

- Superior Low-light and IR Performance
- 8-bit/10-bit MIPI, 1/2/4-lane MIPI
- Automatic Black Level Calibration (ABLC)
- Horizontal and Vertical Mirroring, Windowing and Pixel Binning
- 5 x 5 Statistics Engine for On-chip Auto Exposure Control for Any Programmable Region of Interest (ROI)
- Flexible Control for Row and Column Skip Mode
- On-chip Trigger Mode for Synchronization
- Built in Flash Control
- Two On Chip Phase Lock Loop (PLL)
- Context Switching
- 1056 Bytes One-time Programmable Memory (OTPM)
- Simple Two-wire Fast-mode + Serial Interface

Applications

- Bar Code Scanner
- Factory Automation
- Autonomous Mobile Robot (AMR)
- Machine Vision
- 3D Scanning
- Biometrics

ORDERING INFORMATION

Table 2. AVAILABLE PART NUMBERS

Part Number	Product Description	Product Attribute Description
AR0235CSSM00SMKA0-CP	Mono, 0° CRA	CSP with Protective Film
AR0235CSSM00SMKA0-CP2	Mono, 0° CRA	CSP with Protective Film, MOQ 50 Pieces
AR0235CSSM00SMKA0-CR	Mono, 0° CRA	CSP without Protective Film
AR0235CSSM00SMKAH3-GEVB	Mono, 0° CRA	Demo Board
AR0235CSSM28SMKA0-CP	Mono, 28° CRA	CSP with Protective Film
AR0235CSSM28SMKA0-CP2	Mono, 28° CRA	CSP Chip Tray with Protective Film, MOQ 50 Pieces
AR0235CSSM28SMKA0-CR	Mono, 28° CRA	CSP without Protective Film
AR0235CSSM28SMKAH3-GEVB	Mono, 28° CRA	Demo Board
PRISM1M-AR0235CSSM130110-GEVB	Mono, 28° CRA	Premier Reference Image Sensor Module (PRISM)
AR0235CS1C00SMKA0-CP-E	RGB, 0° CRA	CSP, with Protective Film, Sample
AR0235CSSC00SMKA0-CP	RGB, 0° CRA	CSP, with Protective Film
AR0235CSSC00SMKA0-CP2	RGB, 0° CRA	CSP, with Protective Film, MOQ 50
AR0235CSSC00SMKA0-CR	RGB, 0° CRA	CSP, without Protective Film
AR0235CSSC00SMKAH3-GEVB	RGB, 0° CRA	Demo Board
AR0235CS1C28SMKA0-CP-E	RGB, 28° CRA	CSP, with Protective Film, Sample
AR0235CSSC28SMKA0-CP	RGB, 28° CRA	CSP, with Protective Film
AR0235CSSC28SMKA0-CP2	RGB, 28° CRA	CSP, with Protective Film, MOQ 50
AR0235CSSC28SMKA0-CR	RGB, 28° CRA	CSP, without Protective Film
AR0235CSSC28SMKAH3-GEVB	RGB, 28° CRA	Demo Board
PRISM1M-AR0235CSSC130110-GEVB	RGB, 28° CRA	Premier Reference Image Sensor Module (PRISM)

NOTE: Refer to AR0235 Die Data Sheet for Die Part Numbers and Ordering Information.

Table 3. FRAME RATE OF DIFFERENT MODES OF OPERATION

Resolution	Mode	Frame Rate (frames per sec)
Full_Resolution 1920 x 1200	Master	120
	Slave Integration Start	90
	Slave Integration Time	90
	Slave Integration Start and Readout Start	90
	Slave Integration Start and Integration	depends on exp (not constant frame rate)
2x2_Subsampling 960 x 600	Master	245
	Slave Integration Start	160
	Slave Integration Time	160
	Slave Integration Start and Readout Start	160
	Slave Integration Start and Integration	depends on exp (not constant frame rate)

GENERAL DESCRIPTION

The **onsemi** AR0235CS can be operated in its default mode or programmed for frame size, exposure, gain, and other parameters. The default mode output is a full-resolution image at 120 frames per second (fps). It outputs 10-bit raw data, using the serial (MIPI) output ports. The device may be operated in video (master) mode or in frame trigger mode.

The AR0235CS includes additional features to allow application-specific tuning: windowing, adjustable auto-exposure control, row-skip and column-skip modes and binning modes.

The sensor is designed to operate in a wide temperature range (–30 °C to +85 °C junction).

FUNCTIONAL OVERVIEW

The AR0235CS is a global shutter sensor that generates a stream of pixel data at a constant frame rate. It uses two on-chip, phase-locked loop (PLL) that are enabled to generate all internal clocks from a single master input clocks running between 10 and 48 MHz. The maximum serial output data rate is 850 Mbps/lane. Figure 1 shows a block diagram of the sensor.

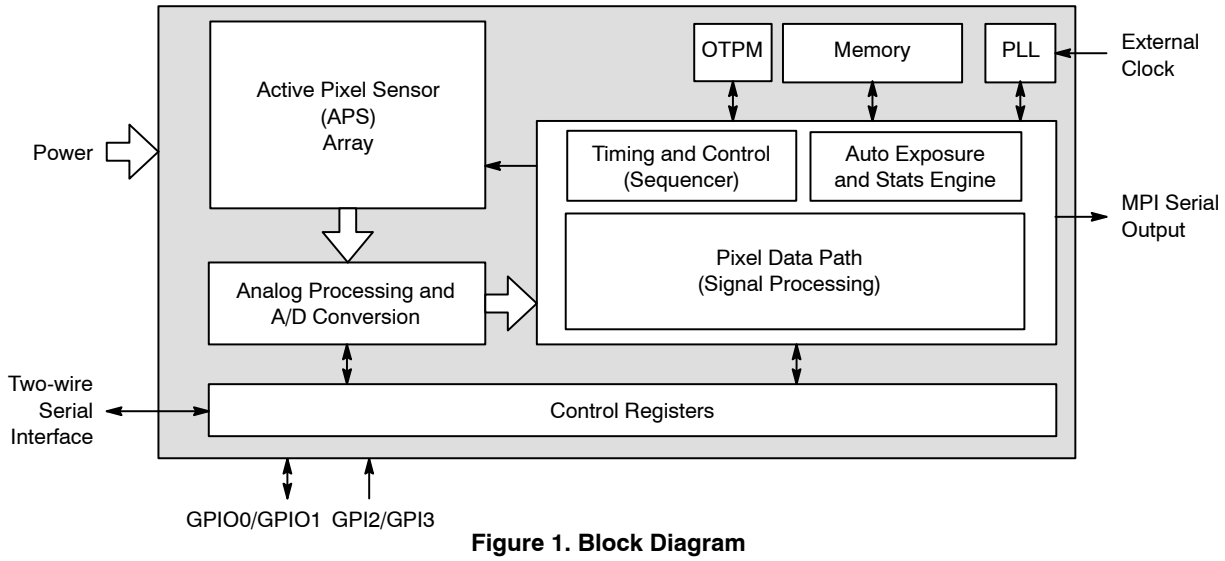


Figure 1. Block Diagram

User interaction with the sensor is through the two-wire serial bus, which communicates with the array control, analog signal chain, and digital signal chain. The AR0235CS features global shutter technology for accurate capture of moving images. The exposure of the entire array is controlled by programming the integration time by register setting. All rows simultaneously integrate light prior to readout. Once a row has been read, the data from the columns is sequenced through an analog signal chain (providing offset correction and gain), and then through an analog-to-digital converter (ADC). The output from the ADC is a 10-bit value for each pixel in the array. The ADC output passes through a digital processing signal chain (which provides further data path corrections and applies digital gain). The pixel data are output at a rate of up to is 850 Mbps/lane, in MIPI to frame and line synchronization signals.

FEATURES OVERVIEW

The AR0235CS Global Shutter sensor has a wide array of features to enhance functionality and to increase versatility. A summary of features follows. Refer to the AR0235CS Developer Guide for a detailed description of features.

- 2.8 μm Global Shutter Pixel
  To improve the low light performance and to capture the moving images accurately a large (2.8 μm) global shutter pixel is implemented for better image optimization.
- Operating Modes
  The AR0235CS works in master (video), and several trigger modes. See Trigger/Slave Mode section for further details.
- Window Control
  Configurable window size and blanking times allow a wide range of resolutions and frame rates. Digital binning and skipping modes are supported, as are vertical and horizontal mirror operations.
- Frame Rate
  AR0235CS is capable of running up to 120 fps at full (1920 x 1200) resolution.
- Embedded Data and Statistics
  The AR0235CS has the capability to output image data and statistics embedded within the frame timing.
- Multi-Camera Synchronization
  The AR0235CS supports advanced line synchronization controls for multi-camera (stereo) support.
- Context Switching and Register Updates
  Context switching may be used to rapidly switch between two sets of register values. The maximum number of contexts is eight.
- Gain
  A programmable analog gain of 1.375x to 7.75x applied globally to all 1920 x 1200 channels is available along with a digital gain of 1x to 8x.
- Automatic Exposure Control
  The integrated automatic exposure control may be used to ensure optimal settings of exposure and gain are computed and updated every other frame. Image can be partitioned into 5 x 5 programmable ROI. Each of these 25 ROI can be selected or deselected for AEC calculation.
- MIPI
  The AR0235CS Global Shutter image sensor supports one, two or four lanes of MIPI data. Compliant to MIPI standards:
  - MIPI Alliance Standard for CSI-2 version 1.2
  - MIPI Alliance Standard for D-PHY version 1.0
- PLL
  Two on chip PLL provides reference clock flexibility and supports spread spectrum sources for improved EMI performance.
- Reset
  The AR0235CS may be reset by a register write, or by a dedicated input pin.
- Black Level Correction
  The AR0235CS sensor has automatic black level correction feature. The magnitude of dark current is measured by utilizing optically dark rows and is used to compensate black level in the visible array.
- Test Patterns
  Several test patterns may be enabled for debug purposes. These include a monochrome bar, PN9, and walking 1s test pattern.
- Silicon/OTPM Revision Information
  A revision register is provided to read out (via I^2C) silicon and OTPM revision information. This will be helpful to distinguish material if there are future OTPM or silicon revisions.
- Compression
  AR0235CS can optionally compress 10-bit data to 8-bit by truncating the least two significant bits.
- Subsampling
  The AR0235CS only performs immediately adjacent pixel binning and skipping. The AR0235CS RGB does not bin same color pixels together as may be desired with a Bayer pattern sensor.
  Binning (horizontal only) is the mean of neighboring two pixels. This will both decrease resolution while keeping the same aspect ratio and improving SNR. In some cases, the frame rate will also increase. Summing (vertical only) is the sum of neighboring two pixels. This will both decrease resolution while keeping the same aspect ratio and improving low light sensitivity.
  The details of the specific binning and summing available will be discussed in the Developer's Guide.
- Gain Control
  The AR0235CS performs global analog and digital gain control. This sensor does not have independent per-channel color gain controls or circuitry.
- Cybersecurity
  AR0235 RGB is a Cybersecurity relevant product for which **onsemi** Cybersecurity Management System (CSMS) processes were followed during development. It is customers responsibility to secure the data through the interfaces and all pins of the device from any unauthorized access. See AR0235CS Developer Guide (AND90094/D) for more details.

PIXEL DATA FORMAT

Default Readout Order

By convention, the sensor core pixel array is shown with the first addressable (logical) pixel (8,8) in the top right corner (see Figure 2). This reflects the actual layout of the array on the die.

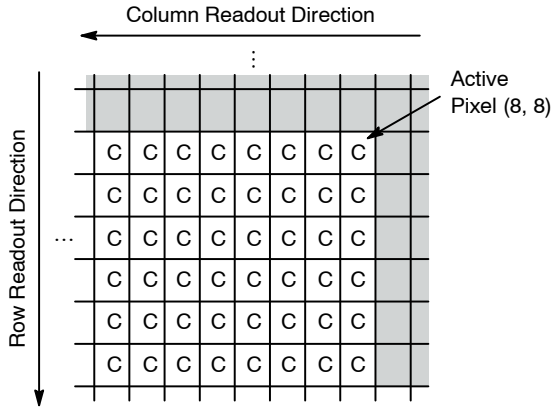


Figure 2. Pixel Mono Pattern Detail (Top Right Corner)

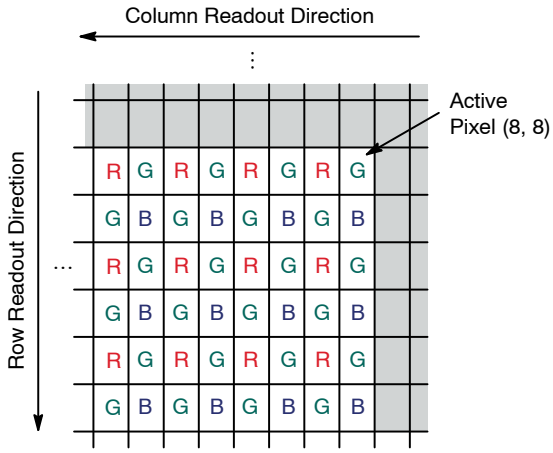
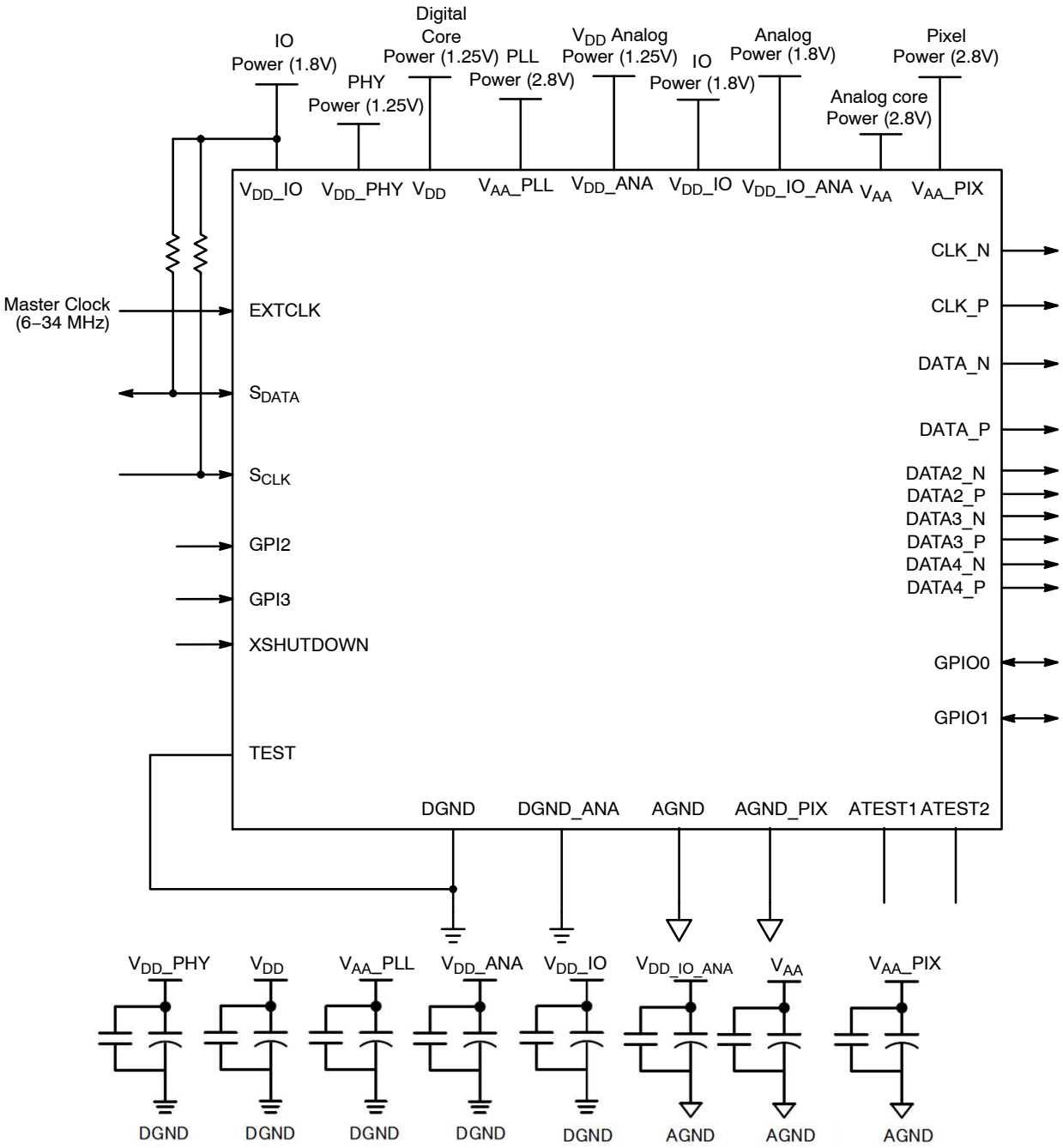


Figure 3. Pixel Color Pattern Detail (Top Right Corner)

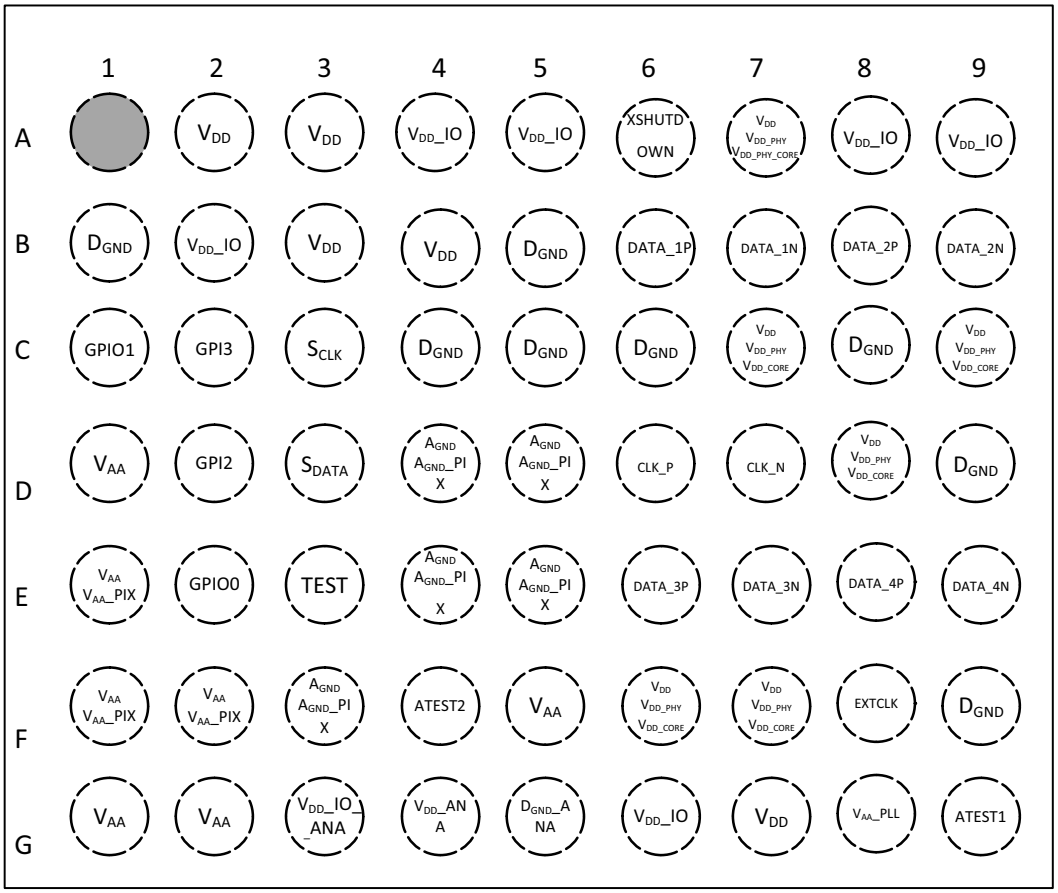
CONFIGURATION AND PINOUT

The figures and tables below show a typical configuration for the AR0235CS image sensor and show the package pinouts.



1. All power supplies must be adequately decoupled.
2. **onsem**i recommends a resistor value of 1.5 kΩ, but a greater value may be used for slower two-wire speed. This pull-up resistor is not required if the controller drives a valid logic level on SCLK at all times.
3. **onsemi** recommends that 0.1 μF and 10 μF decoupling capacitors for each power supply are mounted as close as possible to the pad. Actual values and results may vary depending on the layout and design considerations. Refer to the AR0235CS demo head-board schematic for circuit recommendations.

Figure 4. Serial 4-lane MIPI Interface



Top View
(Ball Down)

Figure 5. 7.66 x 5.64 mm 62-ball CSP Package

Table 4. PIN DESCRIPTIONS – 62-BALL CSP PACKAGE

Name	CSP Ball	Type	Description
DATA_1N	B7	Output	MIPI serial data, lane 1, differential N
DATA_1P	B6	Output	MIPI serial data, lane 1, differential P
CLK_N	D7	Output	MIPI serial clock differential N
CLK_P	D6	Output	MIPI serial clock differential P
DATA2_N	B9	Output	MIPI serial data, lane 2, differential N
DATA2_P	B8	Output	MIPI serial data, lane 2, differential P
DATA3_N	E7	Output	MIPI serial data, lane 3, differential N
DATA3_P	E6	Output	MIPI serial data, lane 3, differential P
DATA4_N	E9	Output	MIPI serial data, lane 4, differential N
DATA4_P	E8	Output	MIPI serial data, lane 4, differential P
VAA	D1, F5, G1, G2	Power	Analog Power (2.8 V)
V_{AA} V_{AA}_PIX	E1, F1, F2	Power	Analog/Pixel Power (2.8 V)
V_{AA}_PLL	G8	Power	PLL Power (2.8 V)
EXTCLK	F8	Input	External input clock
V_{DD} / V_{DD}_PHY / V_{DD}_PHY_CORE	A7, C7, C9, D8, F6, F7	Power	MIPI power (1.25 V)

Table 4. PIN DESCRIPTIONS – 62-BALL CSP PACKAGE (continued)

Name	CSP Ball	Type	Description
D_{GND}	B1, B5, C4, C5, C6, C8, D9, F9	Power	Digital ground
D_{GND}_ANA	G5	Power	Analog ground
AGND/ AGND_PIX	D4,D5,E4,E5,F3	Power	Analog ground, Pixel Ground
V_{DD}	A2, A3, B3, B4, G7	Power	Digital Power (1.25 V)
V_{DD}_ANA	G4	Power	Analog Power (1.25 V)
S_{CLK}	C3	Input	Two-Wire Serial clock input
S_{DATA}	D3	I/O	Two-Wire Serial data I/O
V_{DD}_IO	A4, A5, A8, A9, B2, G6	Power	I/O supply power (1.8 V)
V_{DD}_IO_ANA	G3	Power	I/O supply power (1.8 V)
TEST	E3	Input	Manufacturing test enable pin (connect to DGND)
ATEST1	G9	Input	Analog signal chain manufacturing test access (Leave floating for normal operation).
ATEST2	F4	Input	Analog signal chain manufacturing test access (Leave floating for normal operation).
GPI0[1:0]	C1, E2	I/O	General purpose inputs and outputs. After reset, these pads are not powered-down since its default use is as output. These pads can be configured to provide hardware control of: Output as flash. Input as either SADDR or Trigger. **onsemi** recommends that unused GPIO pins be tied to DGND, but they can also be left floating. If left floating then these pins should not be enabled.
GPI[3:2]	C2, D2	Input	General purpose inputs as SADDR or Trigger. After reset, these pads are powered-down by default; this means that it is not necessary to bond to these pads. **onsemi** recommends that unused GPI pins be tied to DGND, but can also be left floating. If left floating then these pins should not be enabled.
XSHUTDOWN	A6	Input	Asynchronous active LOW reset. This pin will turn off all power domains and is the lowest power state of the sensor. When asserted, data output stops and when deasserted all internal registers are restored to their factory default settings. If XSHUDOWN is not used (floating),this pin must be connected to an external pull-up resistor.
Reserved	A1	N/A	Reserved (do not connect)

Timing Control Modes

The AR0235CS supports advanced line synchronization controls for multi-camera (stereo) support.

Trigger Mode

The slave mode feature of the AR0235CS supports triggering the start of a frame readout from an input signal that is supplied from an external ASIC. The slave mode signal allows for precise control of frame rate and register change updates.

AR0235CS supports the following trigger mode operations.

R0x3CCA	Mode Name
1	Slave Integration Start Mode
2	Slave Integration Time Mode
5	Slave Integration Start and Readout Start Mode
6	Slave Stream Start at Integration Mode

Please refer to the AR0235CS Developer Guide for a detailed description of trigger mode operations.

Trigger/Slave Mode Timing

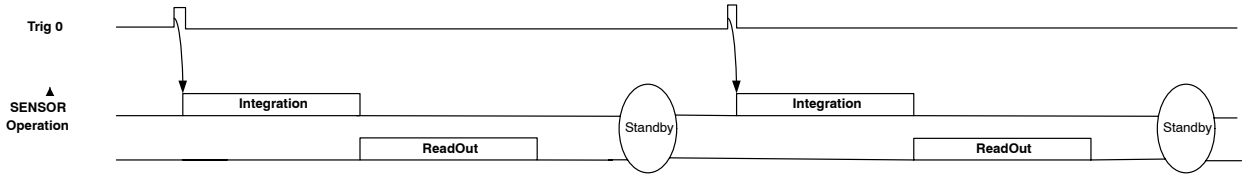


Figure 6. Slave Integration Start Mode

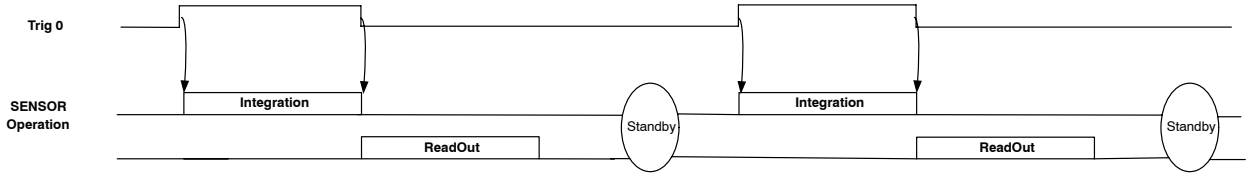


Figure 7. Slave Integration Time Mode

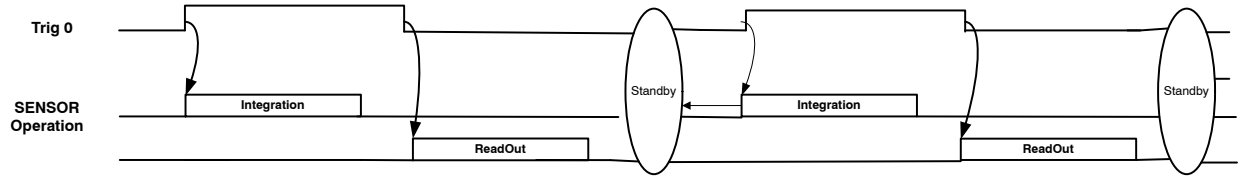


Figure 8. Slave Integration Start and Readout Start Mode

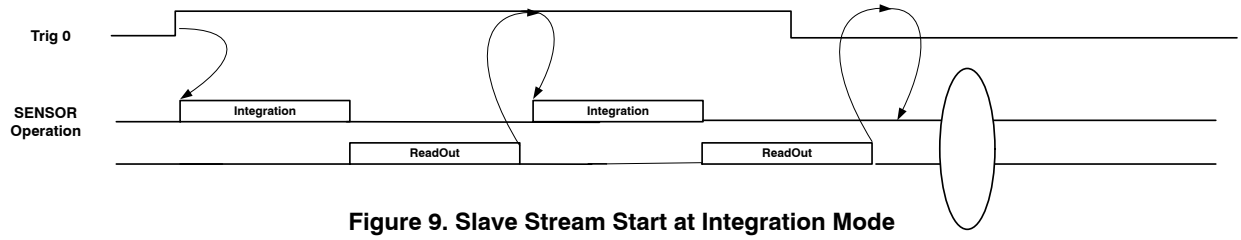


Figure 9. Slave Stream Start at Integration Mode

TWO-WIRE SERIAL REGISTER INTERFACE

The two-wire serial interface bus enables read/write access to control and status registers within the AR0235CS.

The interface protocol uses a master/slave model in which a master controls one or more slave devices. The sensor acts as a slave device. The master generates a clock (S_{CLK}) that is an input to the sensor and is used to synchronize transfers. Data is transferred between the master and the slave on a bidirectional signal (S_{DATA}). S_{DATA} is pulled up to V_{DD_IO} off-chip following I^2C specification. Either the slave or master device can drive S_{DATA} LOW – the interface protocol determines which device is allowed to drive S_{DATA} at any given time.

The protocols described in the two-wire serial interface specification allow the slave device to drive S_{CLK} LOW; the AR0235CS uses S_{CLK} as an input only and therefore never drives it LOW.

Protocol

Data transfers on the two-wire serial interface bus are performed by a sequence of low-level protocol elements:

- a (repeated) start condition
- a slave address/data direction byte
- an (a no) acknowledge bit
- a message byte
- a stop condition

The bus is idle when both S_{CLK} and S_{DATA} are HIGH. Control of the bus is initiated with a start condition, and the bus is released with a stop condition. Only the master can generate the start and stop conditions.

Start Condition

A start condition is defined as a HIGH-to-LOW transition on S_{DATA} while S_{CLK} is HIGH. At the end of a transfer, the master can generate a start condition without previously generating a stop condition; this is known as a “repeated start” or “restart” condition.

Stop Condition

A stop condition is defined as a LOW-to-HIGH transition on S_{DATA} while S_{CLK} is HIGH.

Data Transfer

Data is transferred serially, 8 bits at a time, with the MSB transmitted first. Each byte of data is followed by an acknowledge bit or a no-acknowledge bit. This data transfer mechanism is used for the slave address/data direction byte and for message bytes.

One data bit is transferred during each S_{CLK} clock period. S_{DATA} can change when S_{CLK} is LOW and must be stable while S_{CLK} is HIGH.

Slave Address/Data Direction Byte

Bits [7:1] of this byte represent the device slave address and bit [0] indicates the data transfer direction. A “0” in bit [0] indicates a WRITE, and a “1” indicates a READ. The default slave addresses used by the AR0235CS are 0x6C (write address) and 0x6D (read address) in accordance with the specification. Alternate slave addresses of 0x6E (write address) and 0x6F (read address) can be selected by enabling and asserting the S_{ADDR} input.

An alternate slave address can also be programmed through R0x31FC.

Message Byte

Message bytes are used for sending register addresses and register write data to the slave device and for retrieving register read data.

Acknowledge Bit

Each 8-bit data transfer is followed by an acknowledge bit or a no-acknowledge bit in the S_{CLK} clock period following the data transfer. The transmitter (which is the master when writing, or the slave when reading) releases S_{DATA}. The receiver indicates an acknowledge bit by driving S_{DATA} LOW. As for data transfers, S_{DATA} can change when S_{CLK} is LOW and must be stable while S_{CLK} is HIGH.

No-Acknowledge Bit

The no-acknowledge bit is generated when the receiver does not drive S_{DATA} LOW during the S_{CLK} clock period following a data transfer. A no-acknowledge bit is used to terminate a read sequence.

Typical Sequence

A typical READ or WRITE sequence begins by the master generating a start condition on the bus. After the start condition, the master sends the 8-bit slave address/data direction byte. The last bit indicates whether the request is for a read or a write, where a “0” indicates a write and a “1” indicates a read. If the address matches the address of the slave device, the slave device acknowledges receipt of the address by generating an acknowledge bit on the bus.

If the request was a WRITE, the master then transfers the 16-bit register address to which the WRITE should take place. This transfer takes place as two 8-bit sequences and the slave sends an acknowledge bit after each sequence to indicate that the byte has been received. The master then transfers the data as an 8-bit sequence; the slave sends an acknowledge bit at the end of the sequence. The master stops writing by generating a (re)start or stop condition.

If the request was a READ, the master sends the 8-bit write slave address/data direction byte and 16-bit register address, the same way as with a WRITE request. The master then generates a (re)start condition and the 8-bit read slave address/data direction byte, and clocks out the register data, eight bits at a time. The master generates an acknowledge bit after each 8-bit transfer. The slave’s internal register address is automatically incremented after every 8 bits are transferred. The data transfer is stopped when the master sends a no-acknowledge bit.

Single READ from Random Location

This sequence (Figure 10) starts with a dummy WRITE to the 16-bit address that is to be used for the READ. The

master terminates the WRITE by generating a restart condition. The master then sends the 8-bit read slave address/data direction byte and clocks out one byte of register data. The master terminates the READ by generating a no-acknowledge bit followed by a stop condition. Figure 10 shows how the internal register address maintained by the AR0235CS is loaded and incremented as the sequence proceeds.

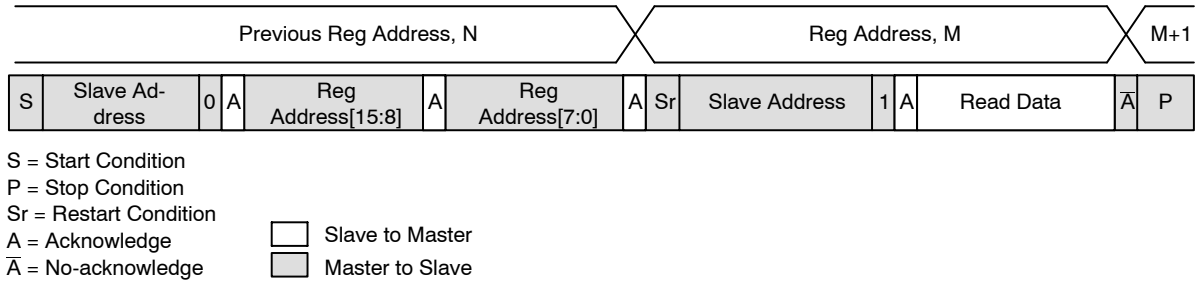


Figure 10. Single READ from Random Location

Single READ from Current Location

This sequence (Figure 11) performs a read using the current value of the AR0235CS internal register address. The master terminates the READ by generating a no-acknowledge bit followed by a stop condition. The figure shows two independent READ sequences.

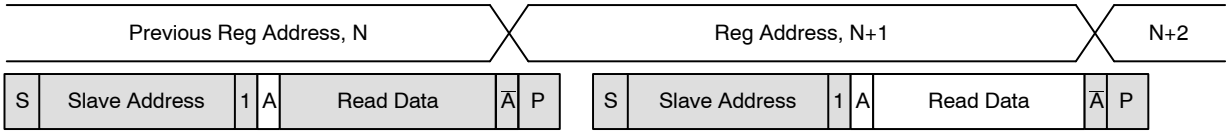


Figure 11. Single READ from Current Location

Sequential READ, Start from Random Location

This sequence (Figure 12) starts in the same way as the single READ from random location (Figure 10). Instead of generating a no-acknowledge bit after the first byte of data has been transferred, the master generates an acknowledge bit and continues to perform byte READs until “L” bytes have been read.

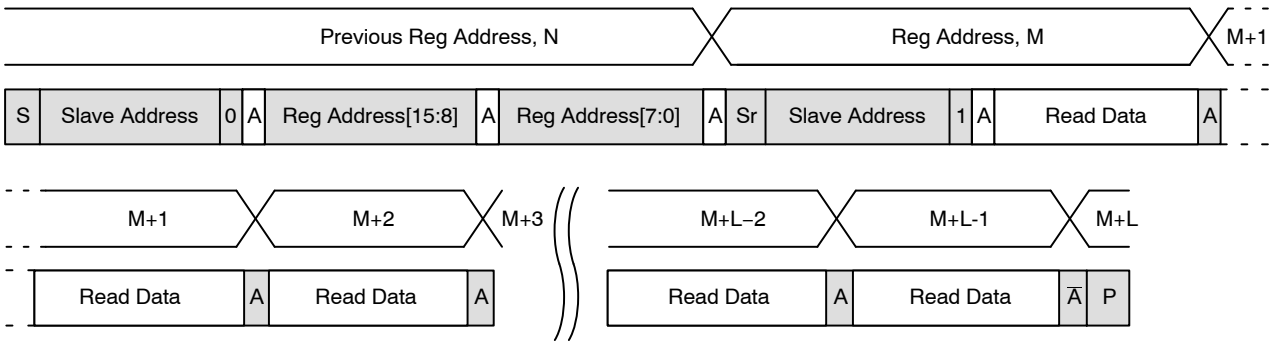


Figure 12. Sequential READ, Start from Random Location

Sequential READ, Start from Current Location

This sequence (Figure 13) starts in the same way as the single READ from current location (Figure 11). Instead of generating a no-acknowledge bit after the first byte of data has been transferred, the master generates an acknowledge bit and continues to perform byte READs until “L” bytes have been read.

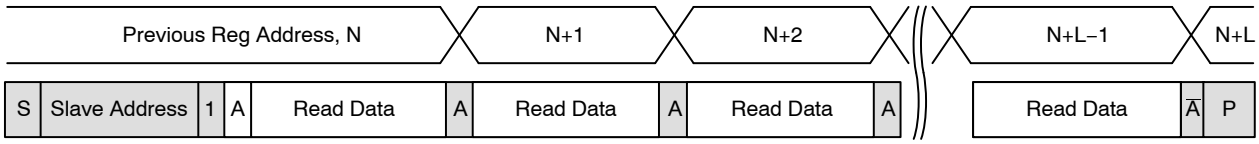


Figure 13. Sequential READ, Start from Current Location

Single WRITE to Random Location

This sequence (Figure 14) begins with the master generating a start condition. The slave address/data direction byte signals a WRITE and is followed by the HIGH then LOW bytes of the register address that is to be written. The master follows this with the byte of write data. The WRITE is terminated by the master generating a stop condition.

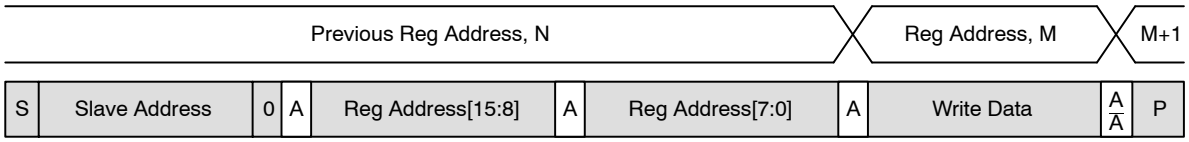


Figure 14. Single WRITE to Random Location

Sequential WRITE, Start at Random Location

This sequence (Figure 15) starts in the same way as the single WRITE to random location (Figure 14). Instead of generating a no-acknowledge bit after the first byte of data has been transferred, the master generates an acknowledge bit and continues to perform byte WRITEs until “L” bytes have been written. The WRITE is terminated by the master generating a stop condition.

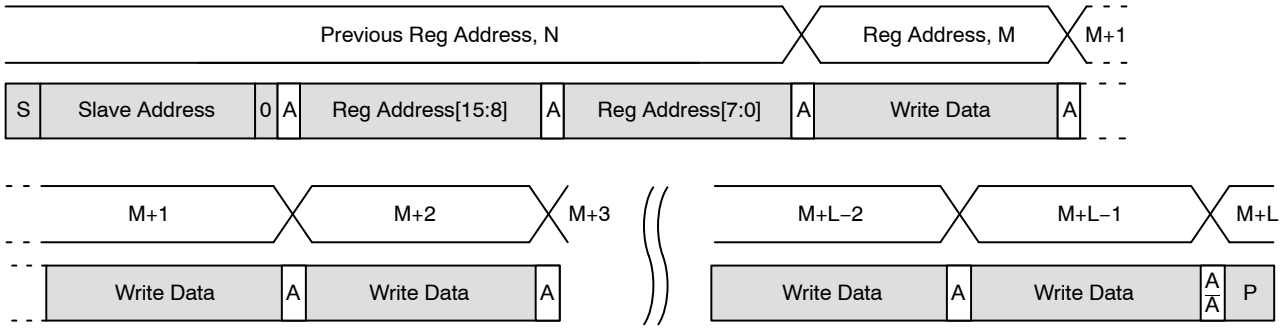


Figure 15. Sequential WRITE, Start at Random Location

SYSTEM STATES

The system states of the AR0235CS are represented as a state diagram in Figure 16 and described in subsequent sections. The effect of XSHUTDOWN on the system state and the configuration of the PLL in the different states are shown in Table 5.

The sensor's operation is broken down into three separate states: hardware standby, software standby, and streaming. The transition between these states might take a certain amount of clock cycles as outlined in Table 5

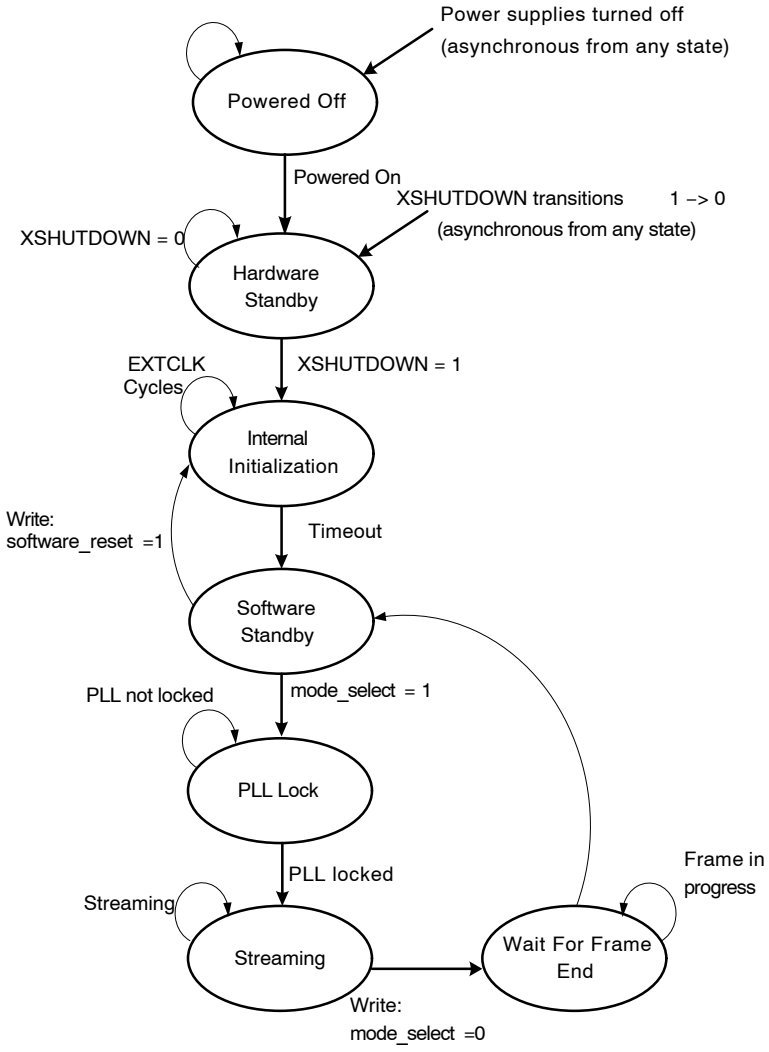


Figure 16. AR0431 System States

Table 5. XSHUTDOWN AND PLL IN SYSTEM STATES

State	XSHUTDOWN	PLL
Powered off	x	VCO powered down
Hardware standby	0	
Internal initialization	1	
Software standby		
PLL Lock		PLL powering up and locking
Streaming		PLL output active
Wait for frame end		

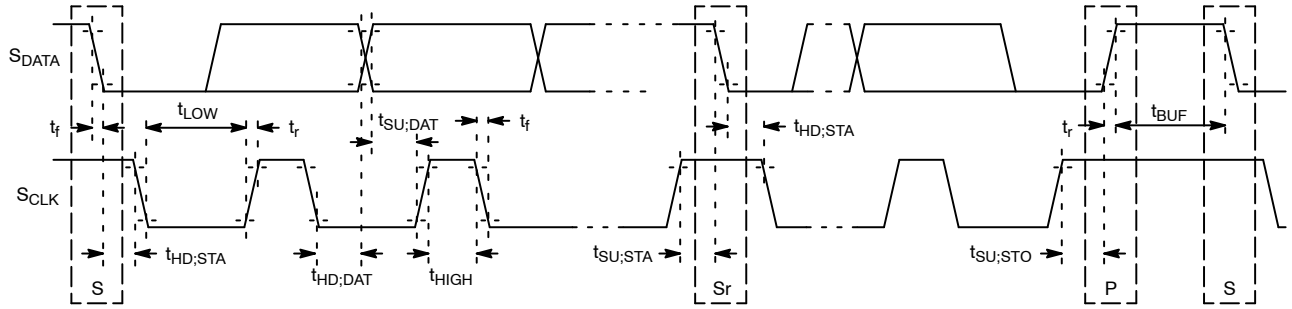
ELECTRICAL SPECIFICATIONS

Unless otherwise stated, the following specifications apply to the following conditions:

V_{DD} = V_{DD}_PHY = 1.25 V ±0.06;
V_{DD}_IO=1.8 V ±0.1 V V_{AA} = V_{AA}_PIX = 2.8 V ±0.2 V;
T_J = −30 °C to +85 °C;
Output Load = 10 pF

Two-Wire Serial Register Interface

The electrical characteristics of the two-wire serial register interface (S_{CLK}, S_{DATA}) are shown in Figure 17 and Table 6.



NOTE: Read sequence: For an 8-bit READ, read waveforms start after WRITE command and register address are issued.

Figure 17. Two-Wire Serial Bus Timing Parameters

Table 6. TWO-WIRE SERIAL BUS CHARACTERISTICS
(f_{EXTCLK} = 25 MHz; V_{DD} = 1.25 V; V_{DD_IO} = 1.8 V; V_{AA} = 2.8 V; V_{AA_PIX} = 2.8 V; V_{DD_PHY} = 1.25 V; T_J = 25 °C)

Parameter	Symbol	Standard Mode		Fast-Mode		Fast-Mode Plus		Unit
		Min	Max	Min	Max	Min	Max	
S_{CLK} Clock Frequency	f_{SCL}	0	100	0	400	0	1000	kHz
Hold Time (Repeated) START Condition	$t_{HD;STA}$	4.0	–	0.6	–	0.26	–	µs
LOW Period of the S_{CLK} Clock	t_{LOW}	4.7	–	1.3	–	0.5	–	µs
HIGH Period of the S_{CLK} Clock	t_{HIGH}	4.0	–	0.6	–	0.26	–	µs
Set-up Time for a Repeated START Condition	$t_{SU;STA}$	4.7	–	0.6	–	0.26	–	µs
Data Hold Time	$t_{HD;DAT}$	0 (Note 5)	3.45 (Note 6)	0 (Note 5)	0.9 (Note 6)	-	–	µs
Data Set-up Time	$t_{SU;DAT}$	250	–	100 (Note 7)	–	0	–	ns
Rise Time of both S_{DATA} and S_{CLK} Signals	t_r	–	1000	20 + 0.1Cb (Note 8)	300	–	120	ns
Fall Time of both S_{DATA} and S_{CLK} Signals	t_f	–	300	20 + 0.1Cb (Note 8)	300	20 x (VDD / 5.5 V)	120	ns
Set-up Time for STOP Condition	$t_{SU;STO}$	4.0	–	0.6	–	0.26	–	µs
Bus Free Time between a STOP and START Condition	t_{BUF}	4.7	–	1.3	–	0.5	–	µs
Capacitive Load for each Bus Line	Cb	–	400	–	400	–	550	pF
Serial Interface Input Pin Capacitance	CIN_SI	–	3.3	–	3.3	-	3.3	pF
S_{DATA} Max Load Capacitance	CLOAD_SD	–	30	–	30	-	0.45	pF
S_{DATA} Pull-up Resistor	Follow I^2C Spec							

2. This table is based on I^2C standard (v2.1 January 2000). Philips Semiconductor.
3. Two-wire control is I^2C-compatible.
4. All values referred to V_{IHmin} = 0.9 V_{DD_IO} and V_{ILmax} = 0.1 V_{DD_IO} levels. Sensor EXCLK = 25 MHz.
5. A device must internally provide a hold time of at least 300 ns for the S_{DATA} signal to bridge the undefined region of the falling edge of S_{CLK}.
6. The maximum $t_{HD;DAT}$ has only to be met if the device does not stretch the LOW period (t_{LOW}) of the S_{CLK} signal.
7. A Fast-mode I^2C-bus device can be used in a Standard-mode I^2C-bus system, but the requirement $t_{SU;DAT}$ 250 ns must then be met. This will automatically be the case if the device does not stretch the LOW period of the S_{CLK} signal. If such a device does stretch the LOW period of the S_{CLK} signal, it must output the next data bit to the S_{DATA} line t_r max + $t_{SU;DAT}$ = 1000 + 250 = 1250 ns (according to the Standard-mode I^2C-bus specification) before the S_{CLK} line is released.
8. Cb = total capacitance of one bus line in pF.

DC Electrical Characteristics

The DC electrical characteristics are shown in Tables 7, 8, and 10.

Table 7. DC ELECTRICAL CHARACTERISTICS

Symbol	Definition	Condition	Min	Typ	Max	Unit
V_{DD}	Core Digital Voltage		1.2	1.25	1.3	V
V_{DD_IO}	I/O Digital Voltage		1.7	1.8	1.9	V
V_{AA}	Analog Voltage		2.6	2.8	3.0	V
V_{AA_PIX}	Pixel Supply Voltage		2.6	2.8	3.0	V
V_{DD_PHY}	MIPI Supply Voltage		1.14	1.25	1.26	V
V_{IH}	Input HIGH Voltage		V_{DD_IO} * 0.7	–	–	V
V_{IL}	Input LOW Voltage		–	–	V_{DD_IO} * 0.3	V
I_{IN}	Input Leakage Current	No Pull-up Resistor; V_{IN} = V_{DD_IO} or D_{GND}	20	–	–	μA
V_{OH}	Output HIGH Voltage		V_{DD_IO} – 0.3	–	–	V
V_{OL}	Output LOW Voltage	V_{DD_IO} = 1.8 V	–	–	0.4	V
I_{OH}	Output HIGH Current	At Specified V_{OH}	−12 (Note 9)	–	–	mA
I_{OL}	Output LOW Current	At Specified V_{OL}	–	–	22	mA

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.
9. A slew rate setting of 7 is needed to achieve I_{OH} and I_{OL} minimum and maximum specifications.

CAUTION: Stresses greater than those listed in Table 8 may cause permanent damage to the device. This is a stress rating only, and functional operation of the device at these or any other conditions above those indicated in the operational sections of this specification is not implied.

Table 8. ABSOLUTE MAXIMUM RATINGS

Symbol	Parameter	Min	Max	Unit
V_{SUPPLY}	Power Supply Voltage (All Supplies)	−0.3	4.5	V
I_{SUPPLY}	Total Power Supply Current	–	200	mA
I_{GND}	Total Ground Current	–	200	mA
V_{IN}	DC Input Voltage	−0.3	V_{DD_IO} + 0.3	V
V_{OUT}	DC Output Voltage	−0.3	V_{DD_IO} + 0.3	V
T_{STG}	Storage Temperature (Note 10)	−40	+150	°C

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.
10. Exposure to absolute maximum rating conditions for extended periods may affect reliability.

Table 9. OPERATING CURRENT CONSUMPTION FOR MIPI OUTPUT

Typ – (V_{AA} = V_{AA_PIX} = 2.8 V; V_{DD_IO} = 1.8 V, V_{DD} = V_{DD_PHY} = 1.25 V); Max MIPI Data Rate 850 Mbps/lane; T_A = 25 °C
Max – (V_{AA} = V_{AA_PIX} = 3 V; V_{DD_IO} = 1.9 V, V_{DD} = V_{DD_PHY} = 1.3 V); Max MIPI Data Rate 850 Mbps/lane; T_A = 60 °C

Symbol	Parameter	Condition	Min	Typ	Max	Unit
I_{DD}	Digital Operating Current	MIPI, Streaming, Full Resolution 120 fps		100	123	mA
I_{DD_IO}	I/O Digital Operating Current			0.1	2.8	
I_{AA}	Analog Operating Current			45	51.5	

11. V_{DD}_PHY is shorted internally in the part. The external supply to V_{DD} and V_{DD}_PHY should be the same supply.

Table 10. STANDBY CURRENT CONSUMPTION

Typ – (V_{AA} = V_{AA_PIX} = 2.8 V; V_{DD_IO} = 1.8 V, V_{DD} = V_{DD_PHY} = 1.25 V); Max MIPI Data Rate 850 Mbps/lane; T_A = 25 °C
Max – (V_{AA} = V_{AA_PIX} = 3 V; V_{DD_IO} = 1.9 V, V_{DD} = V_{DD_PHY} = 1.3 V); Max MIPI Data Rate 850 Mbps/lane; T_A = 60 °C

Symbol	Definition	Condition	Min	Typ	Max	Unit
I_{DD_IO}	I/O Digital Current	STANDBY current when asserting R0x301A[2] = 1(EXTCLK on)		40	49.5	μA
I_{DD}	Digital Current			3480	4690	
I_{AA}/I_{AA_PIX}	Analog/Pixel Current			123	128	
I_{DD_IO}	I/O Digital Current	STANDBY current when asserting R0x301A[2] = 1(EXTCLK off)		18	22.5	μA
I_{DD}	Digital Current			715	1511	
I_{AA}/I_{AA_PIX}	Analog/Pixel Current			120	126	
I_{DD_IO}	I/O Digital Current	XSHUTDOWN signal low (EXTCLK on)		27	33.5	μA
I_{DD}	Digital Current			175	980	
I_{AA}/I_{AA_PIX}	Analog/Pixel Current			1.4	6.6	
I_{DD_IO}	I/O Digital Current	XSHUTDOWN signal low (EXTCLK off)		4.15	5	μA
I_{DD}	Digital Current			170	975	
I_{AA}/I_{AA_PIX}	Analog/Pixel Current			1.2	6.2	

MIPI Electrical Specifications

The **onsemi** AR0235CS sensor supports one, two or four lanes of MIPI data.

Compliant to MIPI standards:

- MIPI Alliance Standard for CSI-2 version 1.0
- MIPI Alliance Standard for D-PHY version 1.2

Table 11. MIPI HIGH-SPEED TRANSMITTER DC CHARACTERISTICS

Symbol	Parameter	Min	Typ	Max	Unit
V_{OD}	HS transmit differential voltage	140	–	270	mV
V_{CMTX}	HS transmit static common mode voltage	150	–	250	mV
ΔV_{OD}	VOD mismatch when output is Differential-1 or Differential-0	–	–	14	mV
$\Delta V_{OMTX(1,0)}$	VCMTX mismatch when output is Differential-1 or Differential-0	–	–	5	mV
V_{OHHS}	HS output HIGH voltage		–	360	mV
Z_{OS}	Single-ended output impedance	40	–	62.5	Ω
ΔZ_{OS}	Single-ended output impedance mismatch	–	–	10	%

Table 12. MIPI HIGH-SPEED TRANSMITTER AC CHARACTERISTICS

Symbol	Parameter	Min	Typ	Max	Unit
	Data bit rate	–	–	900	Mb/s
t_{rise}	20–80% rise time	150	–	333	ps
t_{fall}	20–80% fall time	150	–	333	ps

Table 13. MIPI LOW-POWER TRANSMITTER DC CHARACTERISTICS

Symbol	Parameter	Min	Typ	Max	Unit
V_{OL}	Thevenin output low level	–	–	50	mV
V_{OH}	Thevenin output high level	1.1	1.15	1.3	V
Z_{OLP}	Output impedance of LP transmitter	110	–	–	Ω

Table 14. MIPI LOW-POWER TRANSMITTER AC CHARACTERISTICS

Symbol	Parameter	Min	Typ	Max	Unit
t_{rise}	15–85% rise time	–	–	25	ns
t_{fall}	15–85% fall time	–	–	25	ns
Slew	Slew rate (CLOAD 5–20 pF)	–	–	250	mV/ns
Slew	Slew rate (CLOAD 20–70 pF)	–	–	150	mV/ns

POWER-ON RESET AND STANDBY TIMING

Power-Up Sequence

The recommended power-up sequence for the AR0235CS is shown in Figure 18. The available power supplies (V_{DD_IO}, V_{DD}, V_{DD_PHY}, V_{AA}, V_{AA_PIX}) must have the separation specified below.

1. Set XSHUTDOWN LOW.
2. Power up V_{AA}, V_{AA_PLL} and V_{AA_PIX} (2.8 V).
3. After 100 µs, power up V_{DD_IO} and $V_{DD_IO_ANA}$ (1.8 V).
4. After 100 µs, power up V_{DD}, V_{DD_ANA} and V_{DD_PHY} (1.25 V).
5. After the last power supply is stable, enable EXTCLK.
6. Configure PLL, output, and image settings to desired values.
7. Set streaming mode (R0x301a[2] = 1).

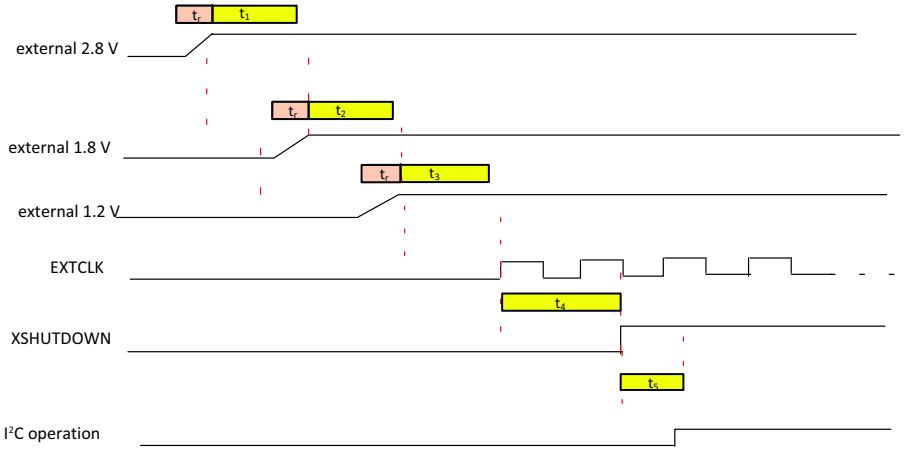


Figure 18. Power Up

NOTE: For CSP package, Supplies having the same voltage (For Example – V_{AA}, V_{AA_PLL} and V_{AA_PIX} (2.8 V)) should be powered on at same time without delay otherwise there will be high leakage current.

Table 15. POWER-UP SEQUENCE

Symbol	Min	Typ	Max	Unit
t1	tr + 100	–	–	µs
t2	tr + 100	–	–	µs
t3	(Note 12)	–	–	
t4	100	–	–	µs
t5	27500	–	–	EXTCLKs
tr	100	–	–	µs

12. Xtal settling time is component-dependent, usually taking about 10–100 ms.
13. Hard reset time is the minimum time required after power rails are settled. In a circuit where hard reset is held down by RC circuit, then the RC time must include the all power rail settle time and Xtal settle time.

Power-Down Sequence

The recommended power-down sequence for the AR0235CS is shown in Figure 19. The available power supplies (V_{DD_IO}, V_{DD}, V_{DD_PHY}, V_{AA}, V_{AA_PIX}) must have the separation specified below.

1. Set software standby mode by setting stream = 0 (R0x301A[2]).
2. Wait till the end of the current frame.
3. Set XSHUTDOWN LOW and clock off.
4. Power off V_{DD}, V_{DD_ANA}, V_{DD_PHY} (1.25 V).
5. After 10 µs, power off V_{DD_IO} and $V_{DD_IO_ANA}$ (1.8 V).
6. After 10 µs, power off V_{AA}, V_{AA_PLL} and V_{AA_PIX} (2.8 V).

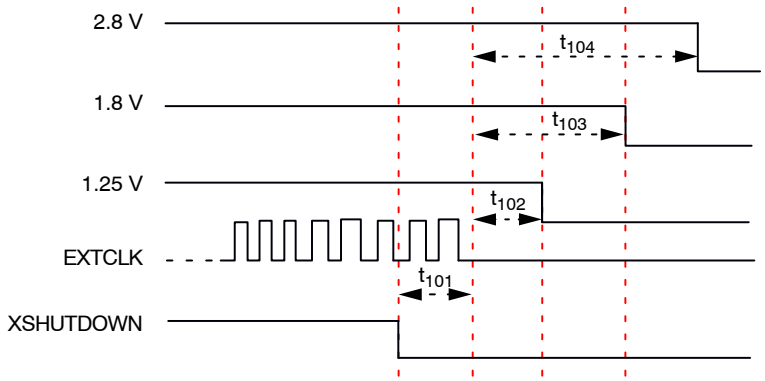


Figure 19. Power Down

NOTE: For CSP package, Supplies having the same voltage (For Example – V_{AA}, V_{AA_PLL} and V_{AA_PIX} (2.8 V)) should be powered off at the same time without delay otherwise there will be high leakage current.

Table 16. POWER-DOWN SEQUENCE

Symbol	Min	Typ	Max
t_{101}	No Limitation	–	10 ms
t_{102}	10 µs	–	–
t_{103}	t_{102} + 10 µs	–	–
t_{104}	max (t_{102}, t_{103}) + 10 µs	–	–

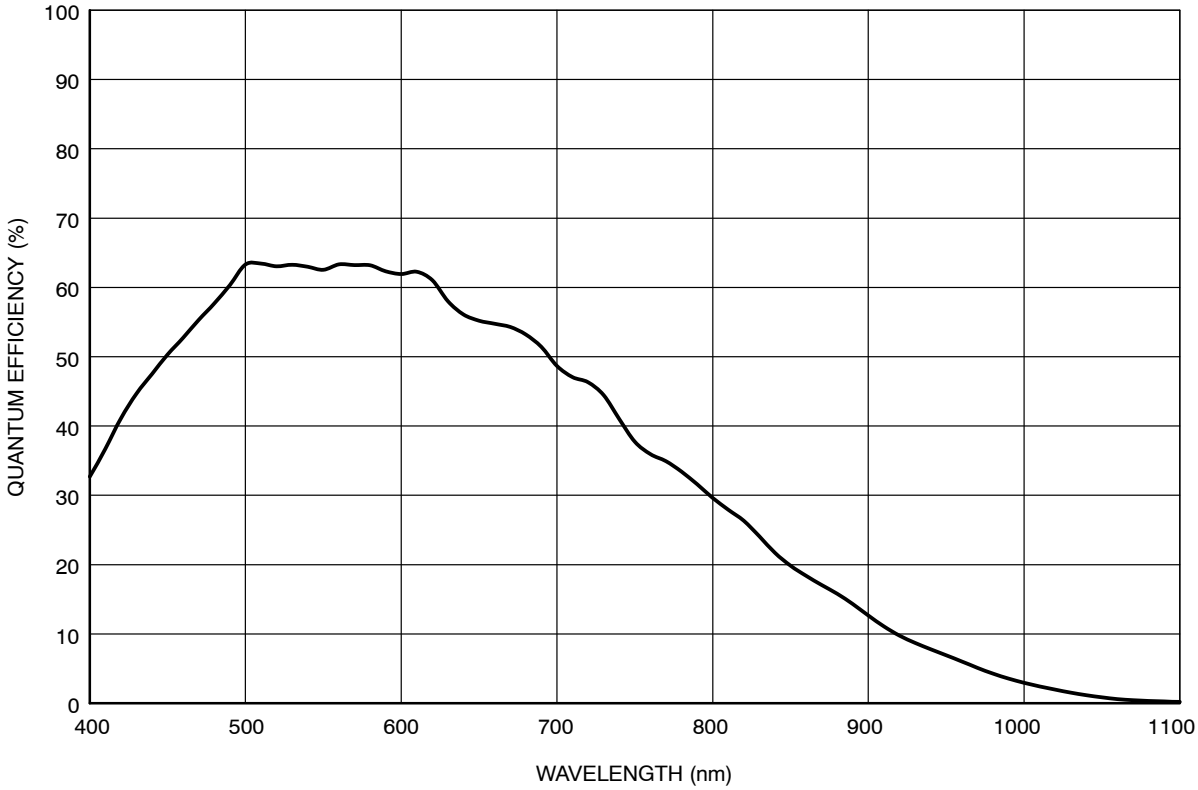


Figure 20. Quantum Efficiency – Monochrome Sensor (Typical)

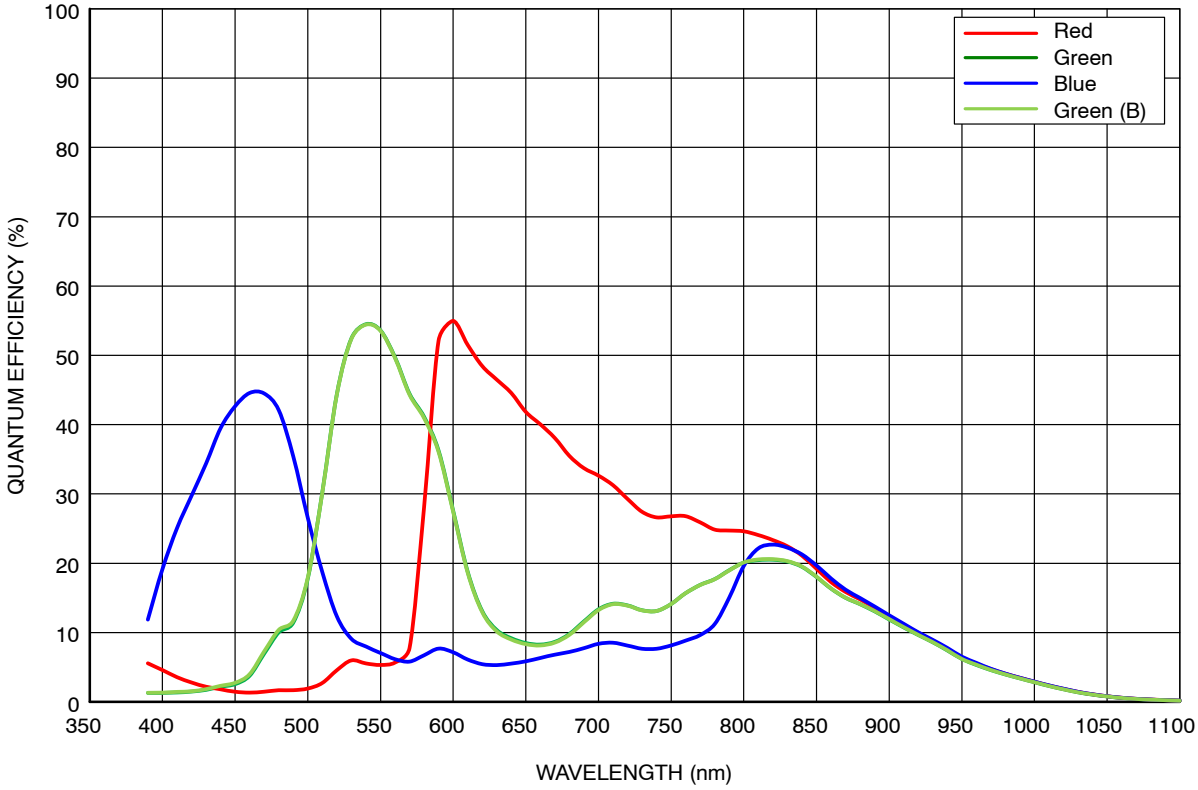


Figure 21. Quantum Efficiency – Color Sensor (Typical)

CRA vs. Image Height Plot

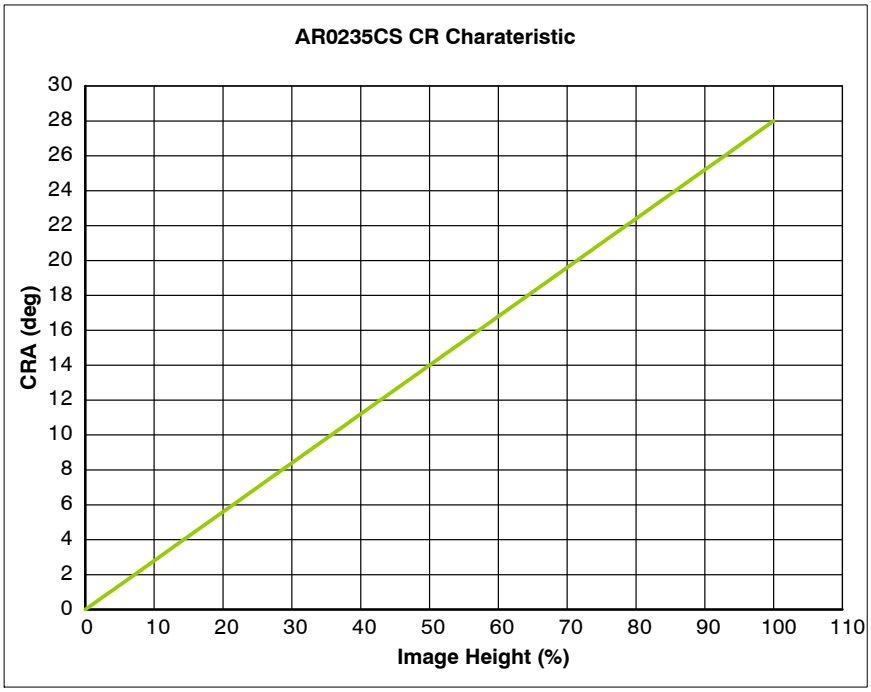


Figure 22. Chief Ray Angle – 28°

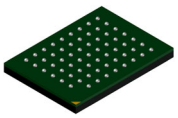
Image Height		CRA
(%)	(mm)	(°)
0	0	0
5	0.158	1.40
10	0.317	2.80
15	0.475	4.20
20	0.634	5.60
25	0.792	7.00
30	0.951	8.40
35	1.109	9.80
40	1.267	11.20
45	1.426	12.60
50	1.584	14.00
55	1.743	15.40
60	1.901	16.80
65	2.060	18.20
70	2.218	19.60
75	2.377	21.00
80	2.535	22.40
85	2.693	23.80
90	2.852	25.20
95	3.010	26.60
100	3.169	28.00

REVISION HISTORY

Revision	Description of Changes	Date
6	Updated context switching bullet in features section.	07/31/2025

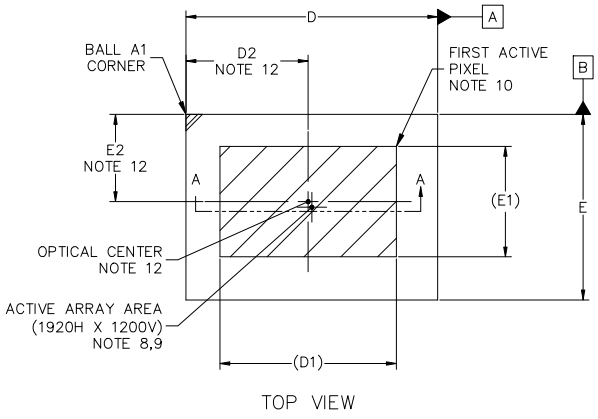
This document has undergone updates prior to the inclusion of this revision history table. The changes tracked here only reflect updates made on the noted approval dates.

PACKAGE DIMENSIONS

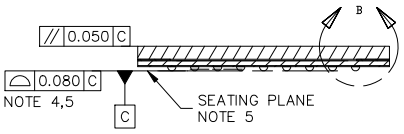


ODCSP62 7.66x5.65x0.62, 0.70P
CASE 570CY
ISSUE B

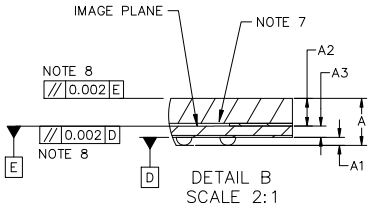
DATE 09 DEC 2024



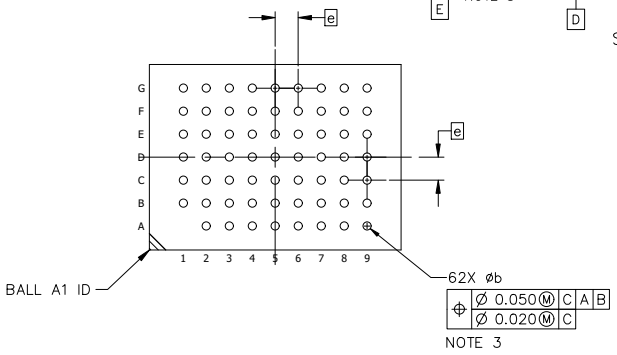
TOP VIEW



SECTION A−A



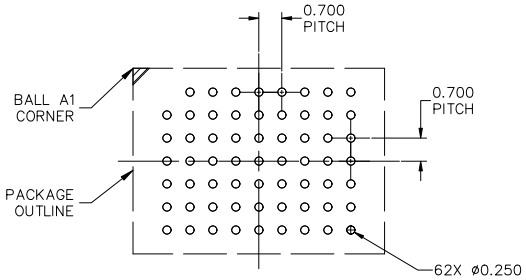
DETAIL B
SCALE 2:1



NOTES:

1. DIMENSIONING AND TOLERANCING PER ASME Y14.5M, 2018.
2. CONTROLLING DIMENSION: MILLIMETERS [mm].
3. SOLDER BALL DIAMETER IS MEASURED AT THE MAXIMUM SOLDER BALL DIAMETER PARALLEL TO DATUM C.
4. COPLANARITY APPLIES TO THE SPHERICAL CROWNS OF THE SOLDER BALLS.
5. DATUM C, THE SEATING PLANE IS DEFINED BY THE SPHERICAL CROWNS OF THE SOLDER BALLS.
6. GLASS: 0.400 THICKNESS; REFRACTIVE INDEX = 1.52.
7. AIR GAP BETWEEN GLASS AND PIXEL ARRAY: 0.045 THICKNESS.
8. PARALLELISM APPLIES ONLY TO THE ACTIVE ARRAY.
9. MAXIMUM ROTATION OF ACTIVE ARRAY RELATIVE TO DATUMS A AND B IS ±0.1°.
10. REFER TO THE DEVICE DATA SHEET FOR TOTAL PIXEL ARRAY DEFINITIONS.
11. PACKAGE CENTER(X, Y) = (0.000, 0.000).
12. OPTICAL CENTER RELATIVE TO PACKAGE CENTER (X, Y) = (−0.114, 0.170).

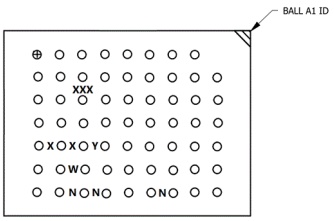
DIM	MILLIMETERS		
	MIN.	NOM.	MAX
A	----	----	0.821
A1	0.085	0.128	0.171
A2	0.430	0.445	0.460
A3	0.138	0.180	0.222
b	0.220	0.250	0.280
D	7.642	7.667	7.692
D1	5.376 REF.		
D2	3.692	3.717	3.742
E	5.626	5.651	5.676
E1	3.360 REF.		
E2	2.631	2.656	2.681
e	0.700 BSC		



RECOMMENDED MOUNTING FOOTPRINT*

*FOR ADDITIONAL INFORMATION ON OUR PB−FREE STRATEGY AND SOLDERING DETAILS, PLEASE DOWNLOAD THE ON SEMICONDUCTOR SOLDERING AND MOUNTING TECHNIQUES REFERENCE MANUAL, SOLDERRM/D.

GENERIC MARKING DIAGRAM*



XXX = Specific Device Code
Y = Year Code
W = Work Week Code
NNN = Serial Number

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, “G” or microdot “ ▪”, may or may not be present. Some products may not follow the Generic Marking.

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